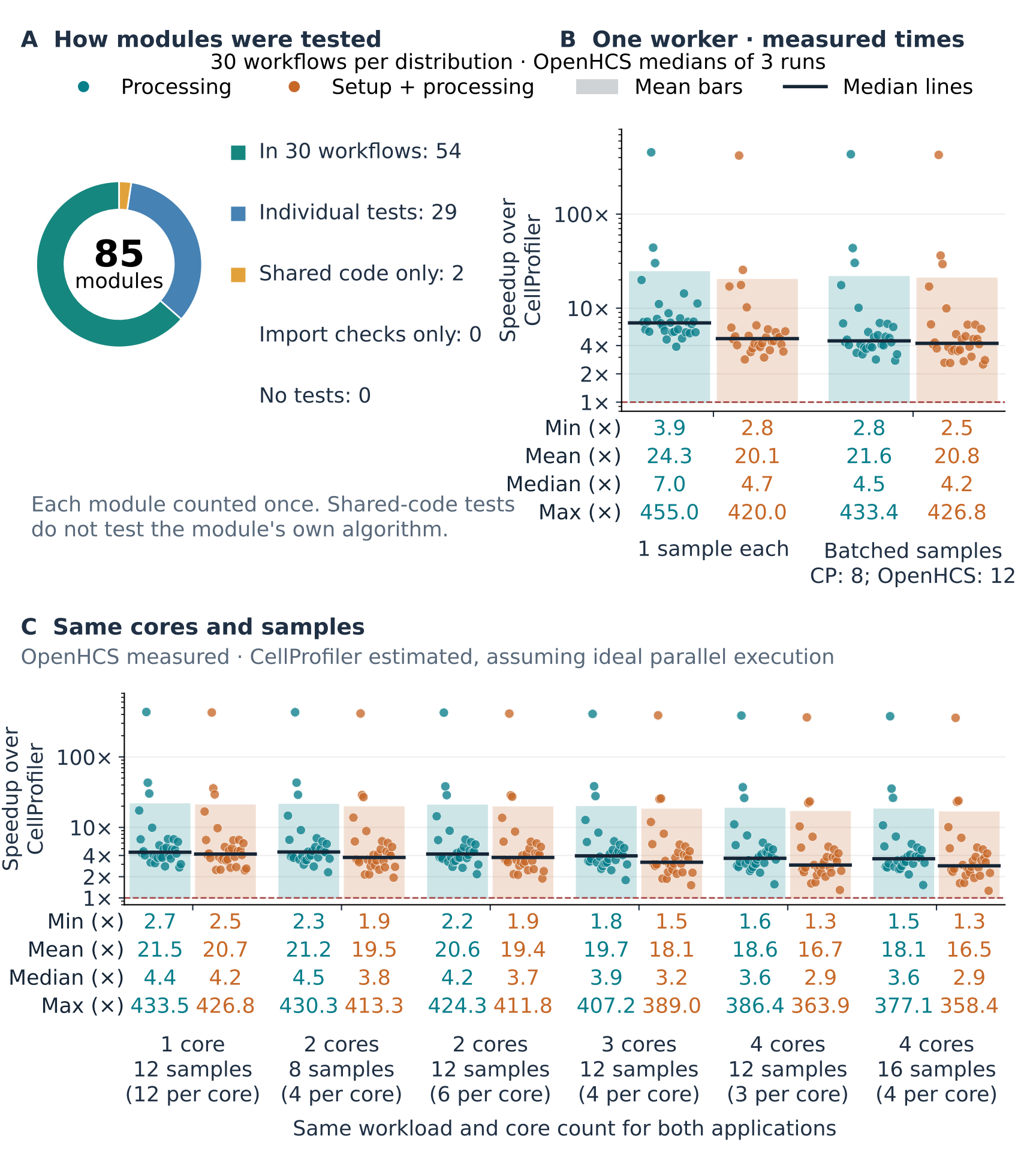
A How modules were tested
B One worker · measured times
30 workflows per distribution · OpenHCS medians of 3 runs
Processing
Setup + processing
Mean bars
Median lines
85
modules
In 30 workflows: 54
Individual tests: 29
Shared code only: 2
Import checks only: 0
No tests: 0
Each module counted once. Shared-code tests do not test the module's own algorithm.
Speedup over CellProfiler
100×
10×
4×
2×
1×
Min (×) 3.9 2.8 2.8 2.5
Mean (×) 24.3 20.1 21.6 20.8
Median (×) 7.0 4.7 4.5 4.2
Max (×) 455.0 420.0 433.4 426.8
1 sample each
Batched samples
CP: 8; OpenHCS: 12
C Same cores and samples
OpenHCS measured · CellProfiler estimated, assuming ideal parallel execution
Speedup over CellProfiler
100×
10×
4×
2×
1×
Min (×) 2.7 2.5 2.3 1.9 2.2 1.9 1.8 1.5 1.6 1.3 1.5 1.3
Mean (×) 21.5 20.7 21.2 19.5 20.6 19.4 19.7 18.1 18.6 16.7 18.1 16.5
Median (×) 4.4 4.2 4.5 3.8 4.2 3.7 3.9 3.2 3.6 2.9 3.6 2.9
Max (×) 433.5 426.8 430.3 413.3 424.3 411.8 407.2 389.0 386.4 363.9 377.1 358.4
1 core 12 samples (12 per core)
2 cores 8 samples (4 per core)
2 cores 12 samples (6 per core)
3 cores 12 samples (4 per core)
4 cores 12 samples (3 per core)
4 cores 16 samples (4 per core)
Same workload and core count for both applications